Configuração 12: $M = 256$, $C = 8$, $K = 12$, IoT = 20 dB, 16-QAM
Perda de SNR do BCD-MMSE (4 iterações) em SER = 10^{-2}, em dB, em relação à referência ideal (float64); × = não chega a 10^{-2} nas SNRs simuladas
Fundo laranja no rótulo: o modo com o melhor resultado no par, comparando do formato com menos bits para o de mais, sem contar as colunas em que todos os modos ficam abaixo de 0,05 dB
Embaixo da perda: maior ocupação da XCU50 entre LUT, DSP, FF e BRAM (síntese HLS, sem diretivas), no mesmo formato e modo

Perda de SNR (dB)

< 0,1 dB | 0,1 a 0,5 dB | 0,5 a 1 dB | 1 a 2 dB | 2 a 3 dB | 3 a 5 dB | 5 a 10 dB | > 10 dB | × não chega a 10^{-2} | rótulo do melhor modo

TDLA30-5

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	4.30 1.8%	0.01 2.1%	0.01 3.1%	0.00 3.6%
	AP_SAT_ZERO	4.30 1.8%	0.01 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	4.30 1.9%	0.01 2.2%	0.01 3.7%	0.00 4.2%
	AP_WRAP	4.30 1.7%	0.01 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	4.30 1.8%	0.01 2.2%	0.01 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	× 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	× 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	× 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	× 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	× 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	1.27 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.27 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.27 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.27 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	1.27 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	1.27 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.27 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.27 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.27 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.27 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	1.27 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.27 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.27 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.27 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	1.27 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	1.27 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.27 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.27 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.27 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.27 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	1.27 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.27 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.27 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.27 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	1.27 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%

TDLA30-75

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	4.34 1.8%	0.00 2.1%	0.00 3.1%	0.00 3.6%
	AP_SAT_ZERO	4.34 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	4.34 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_WRAP	4.34 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	4.34 1.8%	0.00 2.2%	0.00 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	× 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	× 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	× 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	× 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	× 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	1.32 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.32 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.32 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.32 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	1.32 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	1.32 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.32 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.32 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.32 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.32 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	1.34 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.34 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.34 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.34 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	1.34 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	1.34 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.34 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.34 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.34 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.34 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	1.29 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.29 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.29 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.29 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	1.29 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%

TDLA30-300

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	4.22 1.8%	0.01 2.1%	0.01 3.1%	0.00 3.6%
	AP_SAT_ZERO	4.22 1.8%	0.01 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	4.22 1.9%	0.01 2.2%	0.01 3.7%	0.00 4.2%
	AP_WRAP	4.22 1.7%	0.01 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	4.22 1.8%	0.01 2.2%	0.01 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	× 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	× 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	× 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	× 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	× 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	1.29 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.29 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.29 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.29 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	1.29 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	1.27 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.27 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.27 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.27 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.27 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	1.29 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.29 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.29 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.29 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	1.29 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	1.31 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.31 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.31 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.31 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.31 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	1.30 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.30 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.30 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.30 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	1.30 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%

TDLC300-100

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	3.80 1.8%	0.01 2.1%	0.01 3.1%	0.00 3.6%
	AP_SAT_ZERO	3.80 1.8%	0.01 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	3.80 1.9%	0.01 2.2%	0.01 3.7%	0.00 4.2%
	AP_WRAP	3.80 1.7%	0.01 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	3.80 1.8%	0.01 2.2%	0.01 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	× 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	× 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	× 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	× 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	× 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	1.32 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.32 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.32 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.32 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	1.32 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	1.28 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.28 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.28 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.28 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.28 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	1.28 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.28 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.28 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.28 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	1.28 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	1.30 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.30 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.30 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.30 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	1.30 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	1.27 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	1.27 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	1.27 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	1.27 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	1.27 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%

TDLB100-400

Quantização	Estouro	<22,10>	<26,10>	<32,12>	<36,12>
AP_TRN	AP_SAT	1.65 1.7%	0.00 2.0%	0.00 3.3%	0.00 3.7%
	AP_SAT_ZERO	1.65 1.6%	0.00 2.2%	0.00 3.9%	0.00 4.2%
	AP_SAT_SYM	1.65 1.8%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP	1.65 1.6%	0.00 2.1%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	1.65 1.7%	0.17 2.2%	0.00 3.9%	0.00 4.4%
AP_TRN_ZERO	AP_SAT	× 1.9%	0.68 2.3%	0.03 3.9%	0.00 4.4%
	AP_SAT_ZERO	× 1.7%	0.68 2.2%	0.03 3.9%	0.00 4.4%
	AP_SAT_SYM	× 2.1%	0.68 2.5%	0.03 3.9%	0.00 4.3%
	AP_WRAP	× 1.4%	0.68 1.9%	0.03 3.4%	0.00 3.9%
	AP_WRAP_SM	× 1.9%	0.74 2.3%	0.03 3.9%	0.00 4.4%
AP_RND	AP_SAT	1.38 1.8%	0.01 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	1.38 1.7%	0.01 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	1.38 2.0%	0.01 2.4%	0.00 3.9%	0.00 4.4%
	AP_WRAP	1.38 1.6%	0.01 2.1%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	1.38 1.9%	0.20 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_ZERO	AP_SAT	1.39 1.9%	0.01 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	1.39 1.7%	0.01 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	1.39 2.1%	0.01 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	1.39 1.4%	0.01 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	1.39 1.9%	0.20 2.3%	0.00 3.9%	0.00 4.4%
AP_RND_MIN_INF	AP_SAT	1.39 1.8%	0.01 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	1.39 1.7%	0.01 2.2%	0.00 3.9%	0.00 4.2%
	AP_SAT_SYM	1.39 2.0%	0.01 2.4%	0.00 3.9%	0.00 4.3%
	AP_WRAP	1.39 1.6%	0.01 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP_SM	1.40 1.9%	0.20 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_INF	AP_SAT	1.38 1.9%	0.01 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	1.38 1.7%	0.01 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	1.38 2.1%	0.01 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	1.38 1.4%	0.01 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	1.39 1.9%	0.20 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_CONV	AP_SAT	1.39 1.9%	0.01 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	1.39 1.7%	0.01 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	1.39 2.1%	0.01 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	1.39 1.6%	0.01 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP_SM	1.40 1.9%	0.19 2.3%	0.00 3.9%	0.00 4.4%